SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Concept Mapping

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	M.Deepika	**Reg. No.:**	192372296
Section / Batch:	2023	**Date:**	10/07/2026

Code of Conduct

I ______M.Deepika-192372296__________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _______deepika.m__

Instructions

- This test contains 5 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should be with following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	2	20
Search Data Structure	Linear Search and Binary Search	3	20
Recursion, Performance analysis	Recursion	4	20

Tree Data Structures	Classifications of Tree data structure	5	20
	GRAND TOTAL:	100 Marks	

Annexure A – Concept Mapping

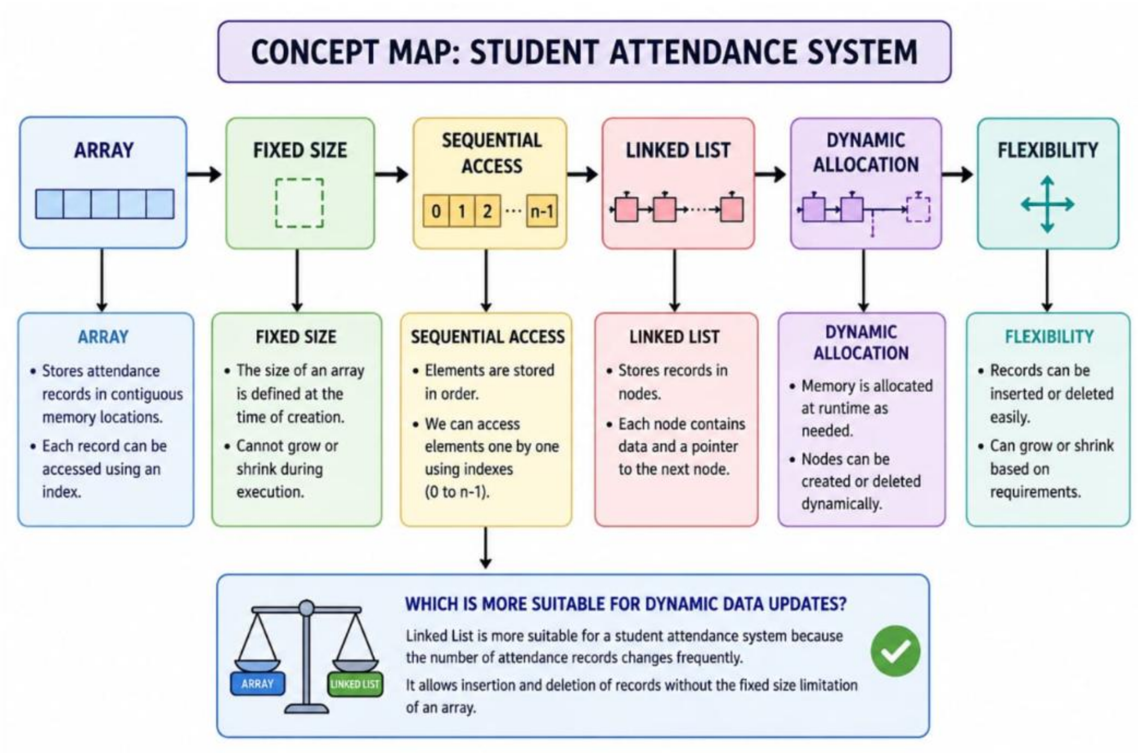
1. A student attendance system records daily presence data. Construct a concept map linking Array → Fixed Size → Sequential Access → Linked List → Dynamic Allocation → Flexibility. Explain how each structure works and justify which is more suitable for dynamic data updates. (20 Marks)
2. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing. (20 Marks)
3. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique. (20 Marks)
4. A program processes nested expressions using recursion and may encounter deep nesting. Construct a concept map linking Recursion → Call Stack → Stack Overflow → Error Handling → Efficiency. Analyze potential issues and justify strategies to handle them. (20 Marks) 5. A company maintains hierarchical employee records with multiple levels of management. Construct a concept map connecting Tree → Nodes → Levels → Traversal (DFS/BFS) → Searching → Efficiency. Analyze how traversal techniques affect performance and justify the most suitable approach for retrieving employee data. (20 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach

Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________



1. ARRAY → FIXED SIZE

An array stores student attendance records in contiguous memory and its size is fixed when created.

ARRAY ↓ **FIXED SIZE**

Example:

Attendance of 5 students for a day

Index:	0	1	2	3	4
Student:	S1	S2	S3	S4	S5
Attendance:	P	A	P	P	A

- Size is fixed (5 in this example).
- If more students join, a new larger array must be created.

2. FIXED SIZE → SEQUENTIAL ACCESS

Array elements can be accessed by index.
Processing attendance records can be done sequentially.

FIXED SIZE ↓ **SEQUENTIAL ACCESS**

Sequential Access in Array

0	1	2	3	4
S1	S2	S3	S4	S5

Access Order:

0 → 1 → 2 → 3 → 4

- Elements are stored in order.
- We can access elements one by one using indexes (0 to n-1).

3. SEQUENTIAL ACCESS → LINKED LIST

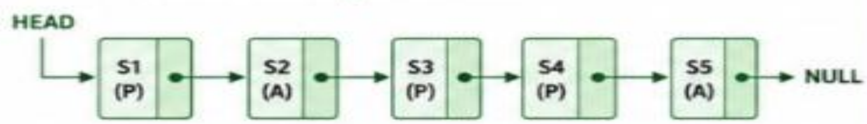
A linked list stores attendance records in nodes.
Each node contains data and a pointer to the next node.

SEQUENTIAL ACCESS

↓

LINKED LIST

Linked List Representation



- Nodes are not stored in contiguous memory.
- We can traverse (access) the list from head to NULL.

4. LINKED LIST → DYNAMIC ALLOCATION → FLEXIBILITY

Linked list nodes are created or deleted during runtime.
This provides flexibility to handle dynamic data.

LINKED LIST → **DYNAMIC ALLOCATION** → **FLEXIBILITY**

DYNAMIC ALLOCATION

- Memory is allocated only when needed.
- Nodes can be created or deleted dynamically.

FLEXIBILITY

- Records can be inserted or deleted easily.
- Can grow or shrink based on requirements.

Which is more suitable for dynamic data updates?

Linked List is more suitable for a student attendance system because the number of attendance records changes frequently. It allows insertion and deletion of records without the fixed size limitation of an array.